

Homework Answer Key: Inverse Trig Functions — Domain, Range & Evaluation

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Part A — Composite Evaluation Using Reference Triangles

Problem 1

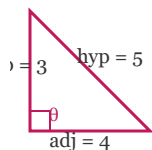
Find the exact value of $\cos\left(\sin^{-1}\left(\frac{3}{5}\right)\right)$.

Reference Triangle

Pythagorean Theorem

Let $\theta = \sin^{-1}\left(\frac{3}{5}\right)$, so $\sin \theta = \frac{3}{5}$.

Build right triangle: opp = 3, hyp = 5 $\rightarrow$ adj = $\sqrt{25 - 9} = \sqrt{16} = 4$.



$$\cos \theta = \frac{\text{adj}}{\text{hyp}} = \frac{4}{5}.$$

$$\frac{4}{5}$$

Problem 2

Find the exact value of $\sin\left(\cos^{-1}\left(\frac{2}{3}\right)\right)$.

Reference Triangle

Let $\theta = \cos^{-1}\left(\frac{2}{3}\right)$, so $\cos \theta = \frac{2}{3}$.

Triangle: $\text{adj} = 2$, $\text{hyp} = 3 \rightarrow \text{opp} = \sqrt{9 - 4} = \sqrt{5}$.

$$\sin \theta = \frac{\sqrt{5}}{3}.$$

$$\frac{\sqrt{5}}{3}$$

Problem 3

Find the exact value of $\tan\left(\sin^{-1}\left(\frac{3}{5}\right)\right)$.

Reference Triangle

Same triangle as Problem 1: $\text{opp} = 3$, $\text{adj} = 4$, $\text{hyp} = 5$.

$$\tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{3}{4}.$$

$$\frac{3}{4}$$

Problem 4

Find the exact value of $\cos\left(\tan^{-1}\left(\frac{4}{3}\right)\right)$.

Reference Triangle

Let $\theta = \tan^{-1}\left(\frac{4}{3}\right)$, so $\tan \theta = \frac{4}{3}$.

Triangle: opp = 4, adj = 3 $\rightarrow$ hyp = $\sqrt{16 + 9} = 5$.

$\cos \theta = \frac{3}{5}$.

$$\frac{3}{5}$$

Problem 5

Write a simplified expression for $\sin\left(\tan^{-1}\left(\frac{x}{2}\right)\right)$ in terms of x .

Variable Triangle

Let $\theta = \tan^{-1}\left(\frac{x}{2}\right)$, so $\tan \theta = \frac{x}{2}$.

Triangle: opp = x , adj = 2 $\rightarrow$ hyp = $\sqrt{x^2 + 4}$.

$$\sin \theta = \frac{x}{\sqrt{x^2 + 4}}.$$

$$\frac{x}{\sqrt{x^2 + 4}}$$

Problem 6

Find the exact value of $\sec\left(\cos^{-1}\left(\frac{1}{4}\right)\right)$.

Reciprocal Trig

Let $\theta = \cos^{-1}\left(\frac{1}{4}\right)$, so $\cos \theta = \frac{1}{4}$.

$$\sec \theta = \frac{1}{\cos \theta} = \frac{1}{1/4} = 4.$$

$$4$$

Part B — Range-Aware Reduction

Problem 7

Evaluate $\sin^{-1}\left(\sin\left(\frac{\pi}{3}\right)\right)$.

Range Check

Range of $\sin^{-1}$ is $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

$\frac{\pi}{3} \approx 1.047 \in [-1.571, 1.571] \rightarrow$ inside range.

Therefore $\sin^{-1}(\sin(\frac{\pi}{3})) = \frac{\pi}{3}$.

$$\frac{\pi}{3}$$

Problem 8

Evaluate $\sin^{-1}\left(\sin\left(\frac{2\pi}{3}\right)\right)$.

Reduction Required

Range Check

$\frac{2\pi}{3} \approx 2.094$. Range of $\sin^{-1}$ is $[-\frac{\pi}{2}, \frac{\pi}{2}] \approx [-1.571, 1.571]$. $\rightarrow$ **outside range**.

Find angle in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ with the same sine value:

$$\sin\left(\frac{2\pi}{3}\right) = \sin\left(\pi - \frac{2\pi}{3}\right) = \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}.$$

$$\text{So } \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3}.$$

$$\frac{\pi}{3}$$

Problem 9

Evaluate $\cos^{-1}\left(\cos\left(\frac{7\pi}{6}\right)\right)$.

Reduction Required

Cosine Symmetry

Range of $\cos^{-1}$ is $[0, \pi]$. But $\frac{7\pi}{6} \approx 3.665 > \pi \rightarrow$ outside range.

Find value: $\cos\left(\frac{7\pi}{6}\right) = -\cos\left(\frac{\pi}{6}\right) = -\frac{\sqrt{3}}{2}$.

$$\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = \pi - \frac{\pi}{6} = \frac{5\pi}{6}.$$

$$\frac{5\pi}{6}$$

Problem 10

Evaluate $\tan^{-1}(\tan(\pi))$.

Reduction Required

Tangent Period

Range of $\tan^{-1}$ is $(-\frac{\pi}{2}, \frac{\pi}{2})$. π is outside this range.

$$\tan(\pi) = 0.$$

$\tan^{-1}(0) = 0$. (The angle inside the range with tangent value 0 is 0.)

$$0$$

Problem 11

Evaluate $\sin^{-1}\left(\cos\left(\frac{3\pi}{2}\right)\right)$.

Co-function

First evaluate the inner function: $\cos\left(\frac{3\pi}{2}\right) = 0$.

Then $\sin^{-1}(0) = 0$, since $0 \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

$$0$$

Part C — Double Angle & Advanced Identities

Problem 12

Find the exact value of $\tan\left(2 \sin^{-1}\left(\frac{1}{3}\right)\right)$.

Double Angle

Reference Triangle

Let $\theta = \sin^{-1}\left(\frac{1}{3}\right)$, so $\sin \theta = \frac{1}{3}$.

Triangle: opp = 1, hyp = 3 $\rightarrow$ adj = $\sqrt{9 - 1} = \sqrt{8} = 2\sqrt{2}$.

$$\tan \theta = \frac{1}{2\sqrt{2}}.$$

Apply double-angle formula: $\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$.

$$\tan^2 \theta = \frac{1}{8}, \text{ so } 1 - \tan^2 \theta = \frac{7}{8}.$$

$$\tan(2\theta) = \frac{2 \cdot \frac{1}{2\sqrt{2}}}{\frac{7}{8}} = \frac{\frac{1}{\sqrt{2}}}{\frac{7}{8}} = \frac{1}{\sqrt{2}} \cdot \frac{8}{7} = \frac{8}{7\sqrt{2}} = \frac{8\sqrt{2}}{14} = \frac{4\sqrt{2}}{7}.$$

$$\frac{4\sqrt{2}}{7}$$

Problem 13

Find the exact value of $\cos\left(2 \tan^{-1}\left(\frac{1}{2}\right)\right)$.

Double Angle

Tangent Form

Let $\theta = \tan^{-1}\left(\frac{1}{2}\right)$, so $\tan \theta = \frac{1}{2}$.

Use: $\cos(2\theta) = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$.

$$\tan^2 \theta = \frac{1}{4}.$$

$$\cos(2\theta) = \frac{1 - \frac{1}{4}}{1 + \frac{1}{4}} = \frac{\frac{3}{4}}{\frac{5}{4}} = \frac{3}{5}.$$

$$\frac{3}{5}$$

Problem 14

Find the exact value of $\sin(\sin^{-1}(0.8) + \cos^{-1}(0.6))$.

Sum Formula

Reference Triangles

Let $\alpha = \sin^{-1}(0.8)$: $\sin \alpha = 0.8 = \frac{4}{5}$, $\cos \alpha = \frac{3}{5} = 0.6$.

Let $\beta = \cos^{-1}(0.6)$: $\cos \beta = 0.6 = \frac{3}{5}$, $\sin \beta = \frac{4}{5} = 0.8$.

Apply $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$:

$$= (0.8)(0.6) + (0.6)(0.8) = 0.48 + 0.48 = 0.96.$$

$$0.96$$

Problem 15

Prove that $\sin^{-1}(x) + \cos^{-1}(x) = \frac{\pi}{2}$ for all $x \in [-1, 1]$.

Identity Proof

Co-function

Proof:

Let $\theta = \sin^{-1}(x)$, where $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Then $\sin \theta = x$, which means $x = \cos\left(\frac{\pi}{2} - \theta\right)$ (co-function identity).

So $\cos^{-1}(x) = \cos^{-1}\left(\cos\left(\frac{\pi}{2} - \theta\right)\right)$.

We need $\frac{\pi}{2} - \theta \in [0, \pi]$ for this to hold directly. Since $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, we have $\frac{\pi}{2} - \theta \in [0, \pi]$. ✓ This is inside the range of $\cos^{-1}$.

Therefore $\cos^{-1}(x) = \frac{\pi}{2} - \theta = \frac{\pi}{2} - \sin^{-1}(x)$.

Rearranging: $\sin^{-1}(x) + \cos^{-1}(x) = \frac{\pi}{2}$. ■

$$\sin^{-1}(x) + \cos^{-1}(x) = \frac{\pi}{2}$$